Hijriyani Nur Riska
23134901S
IA - TNK

No.
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Soal 3 8

1. $q'' + 4q' + 3q = 15 \sin 2t$

Persamaan karakteristik : $m^2 + 4m + 3$

diperoleh akar - akarnya $= (m+1)(m+3)$

$m_1 = -1 \quad \vee \quad m_2 = -3$

penyelesaian PD Homogen adalah $q_h(t) = C_1 e^{-t} + C_2 e^{-3t}$

karena $q(t) = 15 \sin 2t \rightarrow v(t) = a \sin 2t + b \cos 2t$,

$v'(t) = 2a \cos 2t - 2b \sin 2t$,

$v''(t) = -4a \sin 2t - 4b \cos 2t$

substitusikan v, v', dan v'' pada PD yaitu :

$(-4a \sin 2t - 4b \cos 2t) + 4(2a \cos 2t - 2b \sin 2t) + 3(a \sin 2t + b \cos 2t) = 15 \sin 2t$

$(-4a \sin 2t - 4b \cos 2t) + (8a \cos 2t - 8b \sin 2t) + (3a \sin 2t + 3b \cos 2t) = 15 \sin 2t$

$(-a \sin 2t - b \cos 2t) + (8a \cos 2t - 8b \sin 2t) = 15 \sin 2t$

$(-a - 8b) \sin 2t + (-b + 8a) \cos 2t = 15 \sin 2t$

Dengan menyamakan koefisien kedua ruas persamaan, diperoleh

$-a - 8b = 15$

$8a - b = 0 \quad | \times 8$

$-a - 8b = 15$

$64a - 8b = 0$ (-)

$-65a = 15$

$a = -\frac{15}{65} = -\frac{3}{13}$

$-a - 8b = 15$

$\frac{3}{13} - 8b = 15$

$-8b = 15 - \frac{3}{13}$

$b = -\frac{24}{13}$

∴ Solusi umum PDNH $q(t) = C_1 e^{-t} + C_2 e^{-3t} - \frac{3}{13} \sin 2t - \frac{24}{13} \cos 2t$

($q(t)$ stady state)

Jika $t = 0$, $q = 10$, $q' = 0$, maka :

•) $q(0) = C_1 e^{(0)} + C_2 e^{-3(0)} - \frac{3}{13} \sin 2t - \frac{24}{13} \cos 2t$

$10 = C_1 + C_2 - \frac{3}{13} \sin 2(0) - \frac{24}{13} \cos 2(0)$

$10 + \frac{24}{13} = C_1 + C_2$

$\frac{154}{13} = C_1 + C_2 \quad \ldots\ldots\ldots\ldots (1)$

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•) $q'(t) = -C_1 e^{-t} - 3C_2 e^{-3t} - \frac{6}{13}\cos(2t) + \frac{48}{13}\sin(2t)$

$q'(0) = -C_1 e^{-(0)} - 3C_2 e^{-3(0)} - \frac{6}{13}\cos(0) + \frac{48}{13}\sin(0)$

$0 = -C_1 - 3C_2 - \frac{6}{13}$

$\frac{6}{13} = -C_1 - 3C_2 \quad \ldots\ldots\ldots\ldots\ldots\ldots (2)$

Untuk menentukan C_1 dan C_2 gunakan metode eliminasi dari persamaan (1) dan (2)

$C_1 + C_2 = \frac{154}{13}$

$-C_1 - 3C_2 = \frac{6}{13}$ −

$-C_1 - C_2 = -\frac{154}{13}$

$-C_1 - 3C_2 = \frac{6}{13}$ −

$2C_2 = -\frac{160}{13}$

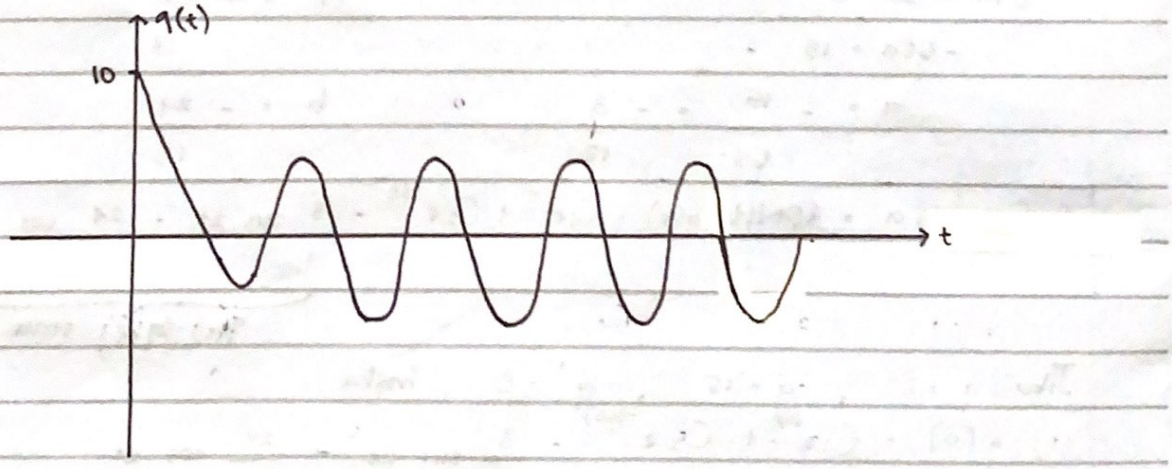
$C_2 = -\frac{80}{13}$

$C_1 + C_2 = \frac{154}{13}$

$-\frac{154}{13} - \frac{80}{13} = -C_1$

$18 = C_1$

∴ Solusi khusus PDNH $q(t) = 18e^{-t} - \frac{80}{13}e^{-3t} - \underbrace{\frac{3}{13}\sin(2t) - \frac{24}{13}\cos(2t)}_{q(t)\text{ stady state}}$



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2. $q'' + 4q' + 4q = 15 \sin 2t$

Persamaan karakteristik $= m^2 + 4m + 4$

diperoleh akar - akarnya $= (m+2)(m+2)$

$m_1 = m_2 = -2$

Penyelesaian PD Homogennya adalah $q_h(t) = e^{-2t}(C_1 + C_2 t)$

karena $q(t) = 15 \sin 2t$, $v = a \sin 2t + b \cos 2t$

$v' = 2a \cos 2t - 2b \sin 2t$

$v'' = -4a \sin 2t - 4b \cos 2t$

substitusikan v, v', dan v'' pada PD yaitu :

$(-4a \sin 2t - 4b \cos 2t) + 4(2a \cos 2t - 2b \sin 2t) + 4(a \sin 2t + b \cos 2t) = 15 \sin 2t$

$(-4a \sin 2t - 4b \cos 2t) + (8a \cos 2t - 8b \sin 2t) + (4a \sin 2t + 4b \cos 2t) = 15 \sin 2t$

$8a \cos 2t - 8b \sin 2t = 15 \sin 2t$

Dengan menyamakan koefisien kedua ruas persamaan, diperoleh :

$8a = 0 \rightarrow a = 0$

$-8b = 15 \rightarrow b = -\frac{15}{8}$

∴ Solusi umum PDNH $q(t) = e^{-2t}(C_1 + C_2 t) \underbrace{- \frac{15}{8} \cos 2t}_{q(t) \text{ stady state}}$

Jika $t = 0$, $q = 10$, $q' = 0$, maka :

•) $q(0) = e^0(C_1 + C_2(0)) - \frac{15}{8} \cos 2(t)$

$10 = C_1 - \frac{15}{8}$

$10 + \frac{15}{8} = C_1$

$\frac{95}{8} = C_1 \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$

•) $q'(t) = -2C_1 e^{-2t} - 2C_2 t e^{-2t} + C_2 e^{-2t} + \frac{15}{4} \sin(2t)$

$q'(0) = -2C_1 e^{-2(0)} - 2C_2(0) e^{-2(0)} + C_2 e^{-2(0)} + \frac{15}{4} \sin(2(0))$

$0 = -2C_1 + C_2 \quad \ldots\ldots\ldots\ldots (2)$

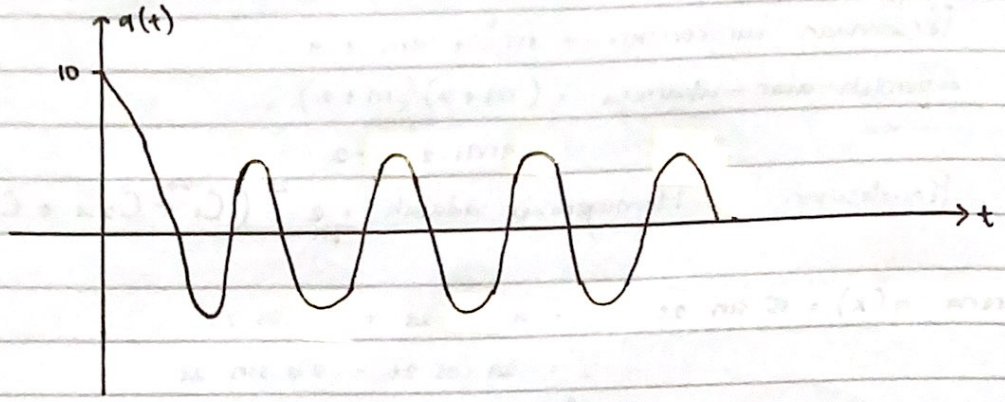
Untuk menentukan C_1 dan C_2 gunakan metode substitusi dari persamaan (1) dan (2)

$C_1 = \frac{95}{8} \rightarrow -2C_1 + C_2 = 0$

$C_2 = \frac{95}{4}$

∴ Solusi khusus PDNH $q(t) = e^{-2t}\left(\frac{95}{8} + \frac{95}{4}t\right) - \frac{15}{8} \cos 2t$

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3. $q'' + 4q' + 5q = 15 \sin 2t$

Persamaan karakteristik : $m^2 + 4m + 5$

diperoleh akar - akarnya : $\dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \dfrac{-4 \pm \sqrt{16 - 20}}{2}$

$= \dfrac{-4 \pm \sqrt{-4}}{2}$

$= -2 \pm j \rightarrow m$

Penyelesaian PD Homogennya adalah $q_h(t) = e^{-2t}(C_1 \cos(t) + C_2 \sin(t))$

karena $q(t) = 15 \sin 2t$, $v = a \sin 2t + b \cos 2t$

$v' = 2a \cos 2t - 2b \sin 2t$

$v' = -4a \sin 2t - 4b \cos 2t$

substitusikan v, v', dan v'' pada PD yaitu :

$(-4a \sin 2t - 4b \cos 2t) + 4(2a \cos 2t - 2b \sin 2t) + 5(a \sin 2t + b \cos 2t) = 15 \sin 2t$

$(-4a \sin 2t - 4b \cos 2t) + (8a \cos 2t - 8b \sin 2t) + (5a \sin 2t + 5b \cos 2t) = 15 \sin 2t$

$(a \sin 2t + b \cos 2t) + (8a \cos 2t - 8b \sin 2t) = 15 \sin 2t$

$(a - 8b) \sin 2t + (b + 8a) \cos 2t = 15 \sin 2t$

Dengan menyamakan koefisien kedua ruas persamaan, diperoleh :

$a - 8b = 15$ ..... (1)

$b + 8a = 0$ ..... (2)

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$a - 8b = 15$

$-8b - 64a = 0$ $-$

$65a = 15$

$a = \frac{3}{13}$

$a - 8b = 15$

$\frac{5}{13} - 8b = 15$

$b = -\frac{29}{13}$

$q(t)$ steady state

∴ Solusi Umum PDNH $q(t) = e^{-2t}(C_1 \cos(t) + C_2 \sin(t)) + \frac{3}{13}\sin(2t) - \frac{29}{13}\cos(2t)$

Jika $t = 0$, $q = 10$, $q' = 0$, maka

→ $q(0) = e^0(C_1 \cos(0) + C_2 \sin(0)) + \frac{3}{13}\sin(0) - \frac{29}{13}\cos(0)$

$10 = C_1 - \frac{29}{13}$

$\frac{154}{13} = C_1$ ..... (1)

→ $q'(t) = -2C_1e^{-2t}\cos(t) - 2C_2e^{-2t}\sin(t) - C_1e^{2t}\sin(t) + C_2e^{-2t}\cos(t) + \frac{3}{13}\cos(2t) + \frac{29}{13}\sin(2t)$

$q'(0) = -2C_1e^0\cos(0) - 2C_2e^0\sin(0) - C_1e^0\sin(0) + C_2e^0\cos(0) + \frac{6}{13}\cos(0) + \frac{48}{13}\sin(0)$

$0 = -2C_1 + C_2 + \frac{6}{13}$

$-\frac{6}{13} = -2C_1 + C_2$ ..... (2)

Untuk menentukan C_1 dan C_2 gunakan metode substitusi dari persamaan (1) dan (2)

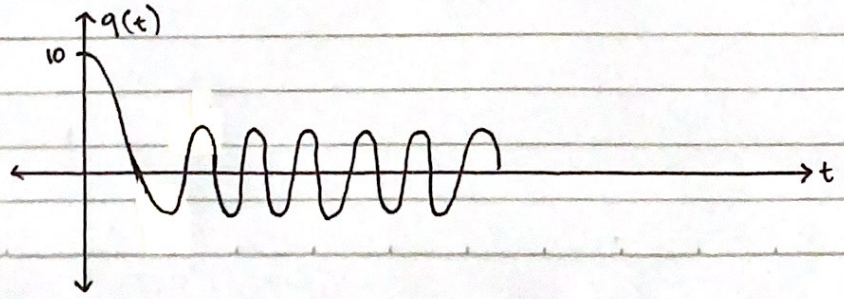
$\frac{154}{13} = C_1$ → $-2C_1 + C_2 = -\frac{3}{13}$

$-2\left(\frac{154}{13}\right) + C_2 = -\frac{6}{13}$

$C_2 = \frac{302}{13}$

$q(t)$ stady state

∴ Solusi khusus PDNH $q(t) = e^{-2t}\left(\frac{154}{13}\cos(t) + \frac{302}{13}\sin(t)\right) + \frac{3}{13}\sin(2t) - \frac{29}{13}\cos(2t)$



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